Four-mode three-node repeater chain

Setup

Message: "Quantum networks will link distant processors into one machine. This message crossed one h..." (447 chars, 3576 bits; 3x the two-node paragraph run). Topology: NodeA --50km-- NodeB --50km-- NodeC on ibm_marrakesh; NodeB is a measure-and-reprepare repeater (trusted-node relay). 256 shots per circuit, 358 circuits per node stage. Link transition probabilities per hop: P(0->1)=0.06502, P(1->0)=0.33161 — numerically identical in all four modes, so every difference between modes is node-error attribution.

The four modes

1. REAL HARDWARE - three batched SamplerV2 jobs on ibm_marrakesh (job ids: d9cqdb2neu4c739mapq0, d9cqekaneu4c739marcg, d9cqg0cjeosc73fgkpr0; billed [27.0, 27.0, 27.0]s). 2. DIGITAL TWIN - Aer with the device-averaged model fitted from all collected calibration snapshots. 3. LAYOUT-AWARE TWIN - Aer with calibration averaged over the qubits the jobs actually run on (trivial layout, physical qubits 0-9, verified from job inputs) with the per-qubit asymmetric readout pair. 4. EQUATIONS - closed-form analytical node model.

Result

Received at node C: hardware CORRUPTED (char accuracy 2.9%). End-to-end raw BER: hardware 30.585%, Digital twin (device avg) 30.934%, Layout-aware twin 30.661%, Equations model 31.132%. Closest model to hardware (summed |delta| over all stages): Layout-aware twin. NOTE: an analog measure-and-reprepare relay copies each bit's degraded distribution instead of regenerating it, so after two 50 km hops the sent-1 bits' P(1) crosses below the 0.5 decode threshold and the message corrupts in EVERY mode, hardware included - the models predict exactly this failure. A decode-and-forward repeater (majority vote at node B, clean re-encode) would regenerate the bits and deliver the message; that variant needs no hardware to simulate.

Provenance (prediction, not fit)

Earliest hardware job submitted 2026-07-17T03:51:08+00:00. Layout twin pinned to snapshot #377 collected 2026-07-17T03:22:43+00:00 (predates jobs: True). Device-avg twin fitted from 130 snapshots, newest 2026-07-17T03:22:43+00:00 (predates jobs: True). The equations model uses no measured data. No simulated mode sees any hardware result.

Calibrations and per-stage BER

Node model parameters

Param	Device-avg twin	Layout twin	Equations
T1 (us)	170.1	212.7	150.0
T2 (us)	104.0	163.3	100.0
1Q gate error	0.000351	0.000280	0.000422
2Q gate error	0.002755	0.002057	0.004663
Readout error	0.011079	0.006030	0.010660
Readout P(1|0)	0.0070537709137709235	0.004045758928571429	None
Readout P(0|1)	0.010918463726884807	0.009228515625	None

Raw per-shot BER by stage

Stage	Hardware	Device-avg twin	Layout	Equations
Node A	0.591%	0.890%	0.643%	1.070%
Link A-B	19.025%	19.206%	19.057%	19.314%
Node B (rep)	19.424%	19.790%	19.492%	20.010%
Link B-C	30.389%	30.609%	30.430%	30.742%
End-to-end	30.585%	30.934%	30.661%	31.132%

Layout

Layout twin qubits: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9] (strategy first_n, matching the trivial placement of the submitted jobs).

Per-channel breakdown (BER contribution, pp)

How to read

Each Aer error channel enabled ALONE at one node stage (links noise-free contribute the baseline 29.949pp end-to-end). node A is measured at stage A; node B / node C are the end-to-end delta vs the baseline. 2q-gate and thermal channels are attached but never triggered by these 1q-only circuits — their rows read the sampling floor (~±0.015 pp at 2048 shots), not signal. Hardware physically cannot be decomposed per channel.

Digital twin (device avg)

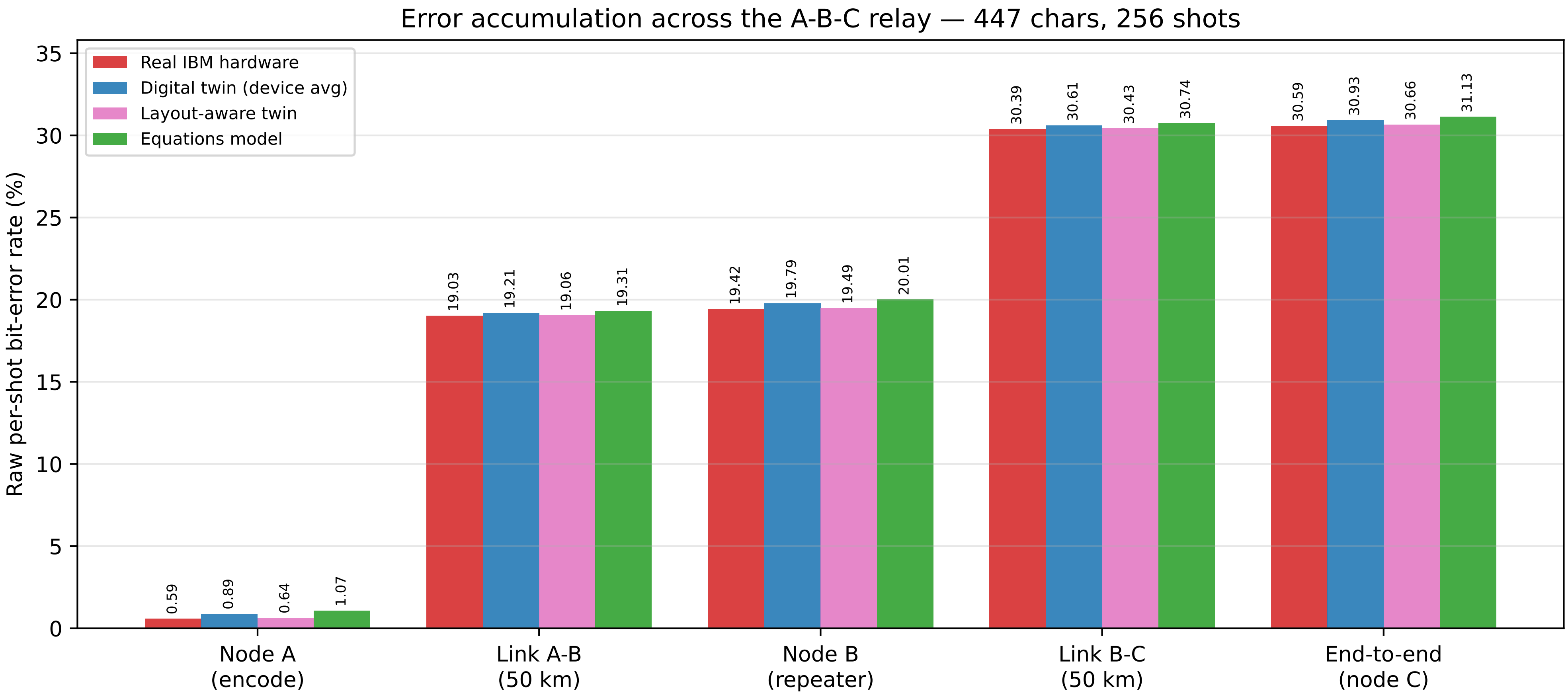
Channel	node A	node B	node C
1-qubit gate	0.0088	0.0047	0.0101
2-qubit gate	0.0000	0.0000	0.0000
Thermal relaxation (T1/T2)	0.0000	0.0000	0.0000
Readout	0.8837	0.3190	0.3379
Full model (per stage)	0.8866	0.3183	0.3484

Layout-aware twin

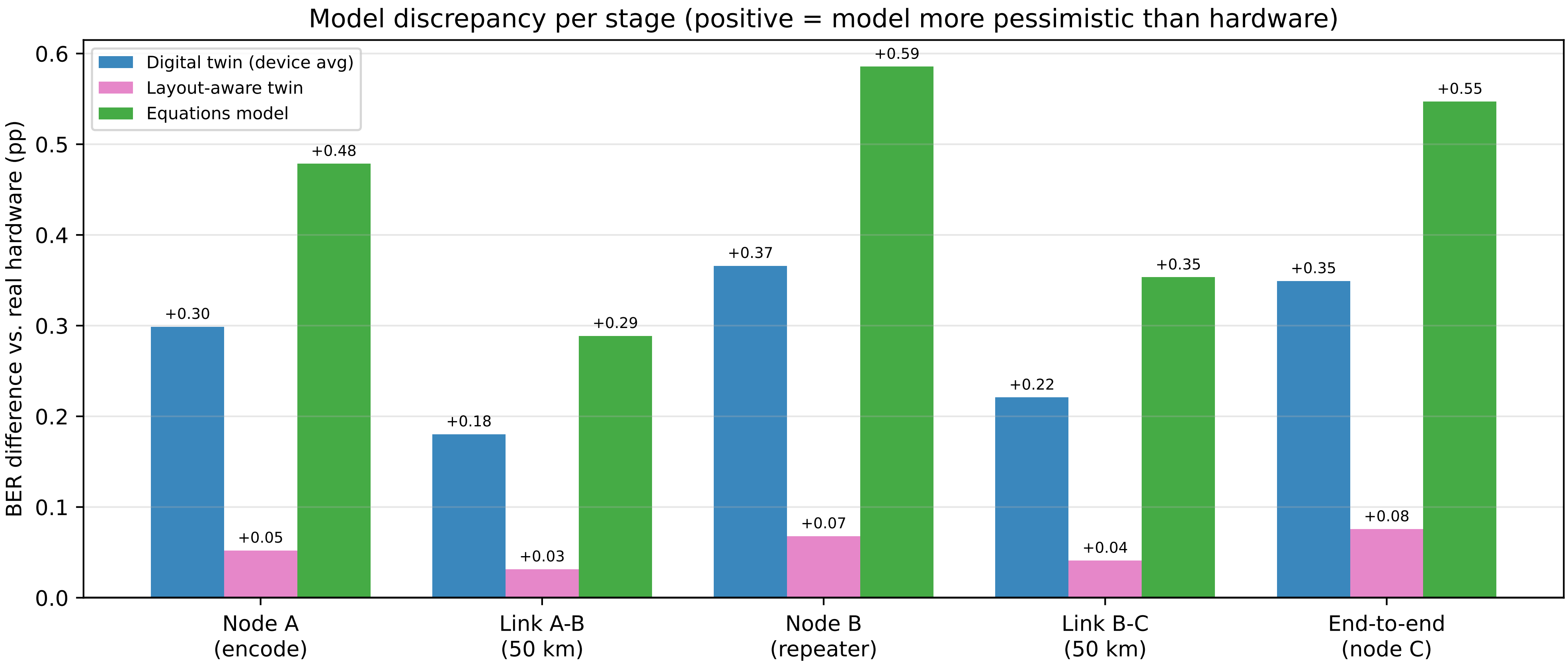
Channel	node A	node B	node C
1-qubit gate	0.0069	0.0054	0.0068
2-qubit gate	0.0000	0.0000	0.0000
Thermal relaxation (T1/T2)	0.0000	0.0000	0.0000
Readout	0.6425	0.2308	0.2373
Full model (per stage)	0.6465	0.2305	0.2444

Equations model

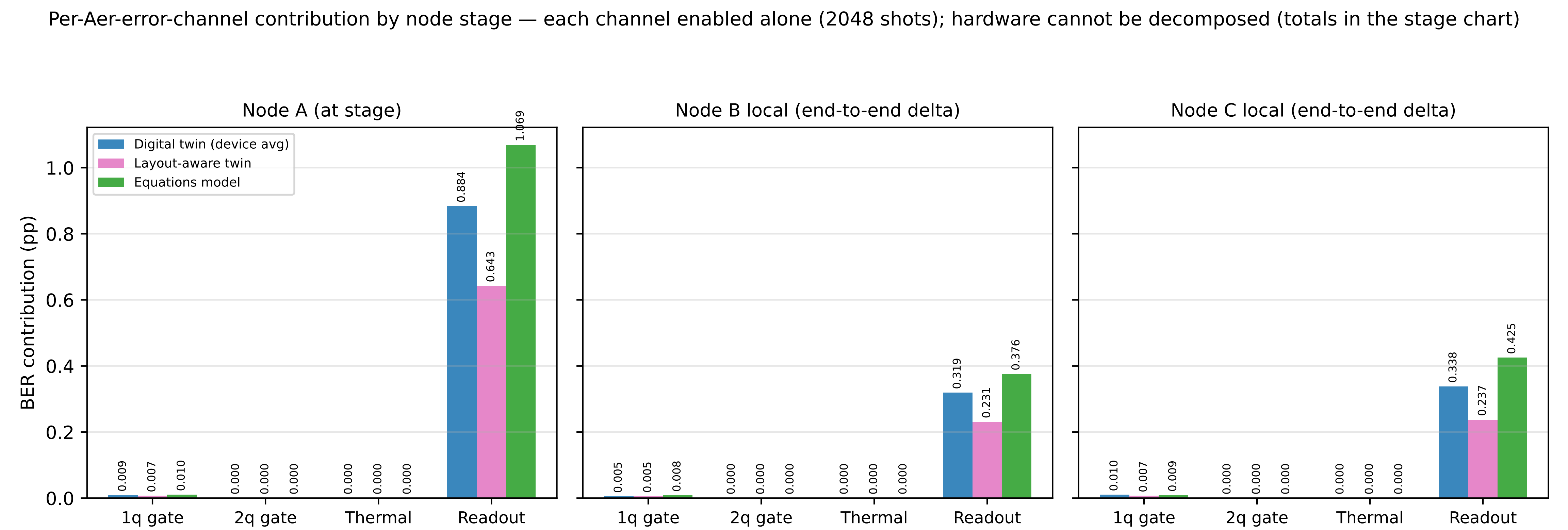
Channel	node A	node B	node C
1-qubit gate	0.0103	0.0082	0.0085
2-qubit gate	0.0000	0.0000	0.0000
Thermal relaxation (T1/T2)	0.0000	0.0000	0.0000
Readout	1.0686	0.3755	0.4250
Full model (per stage)	1.0725	0.3848	0.4337



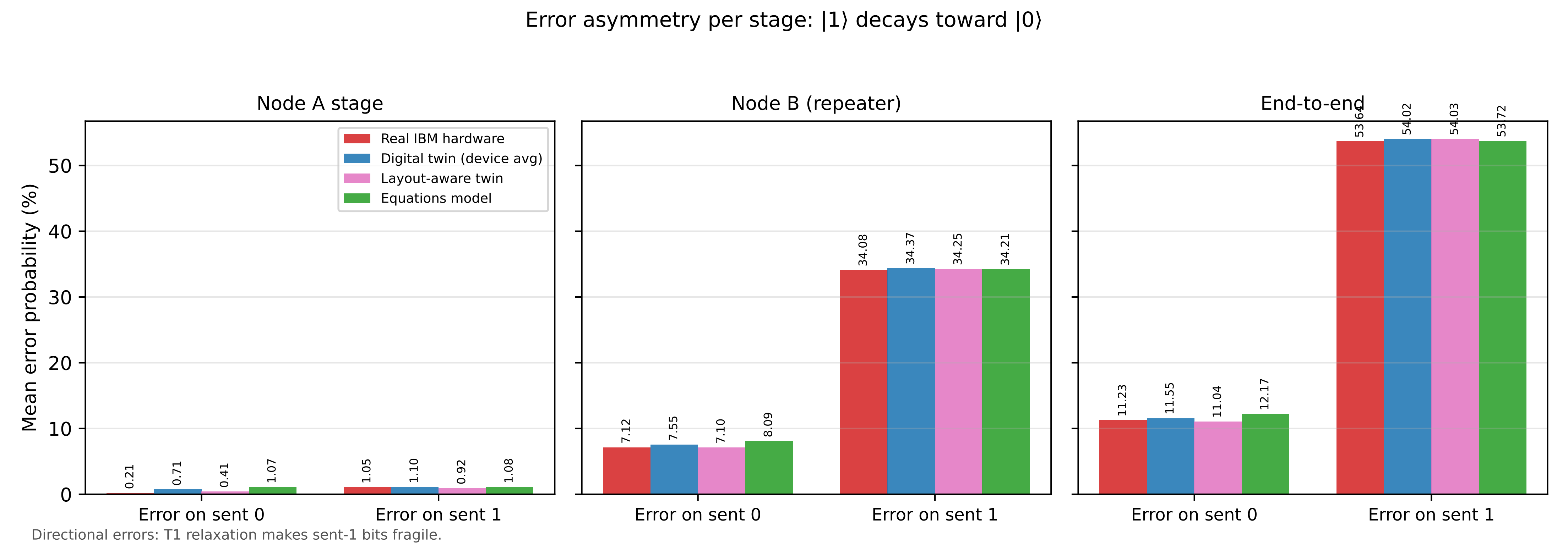
Raw per-shot BER after every stage of the relay.



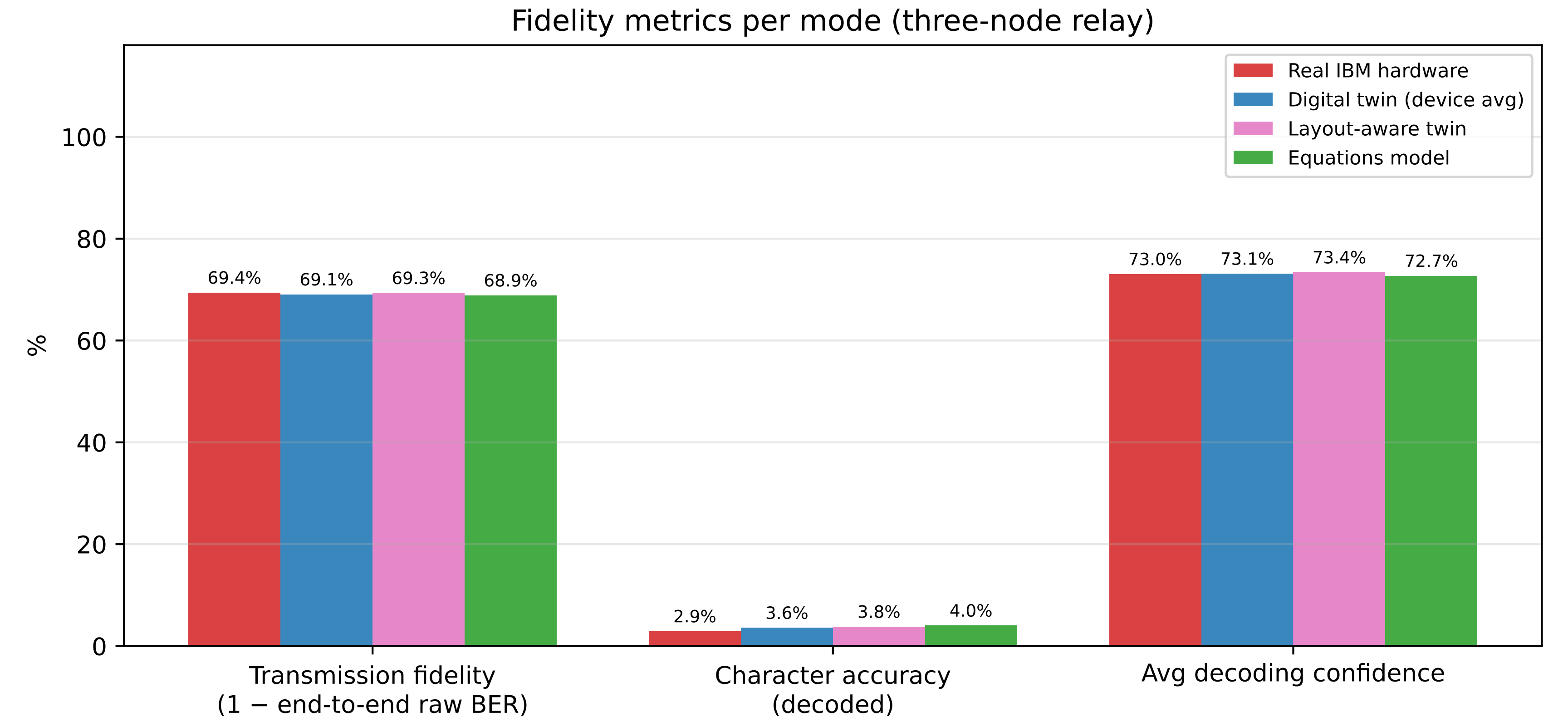
Model minus hardware, per stage.



Per-Aer-channel ablation, per node stage, simulated modes.



Error asymmetry per stage: |1⟩ decays toward |0⟩
Node A stage
Node B (repeater)
End-to-end
Mean error probability (%)
0
10
20
30
40
50
Real IBM hardware
Digital twin (device avg)
Layout-aware twin
Equations model
0.21
0.71
0.41
1.07
1.05
1.10
0.92
1.08
7.12
7.55
7.10
8.09
34.08
34.37
34.25
34.21
11.23
11.55
11.04
12.17
53.64
54.02
54.03
53.72
Error on sent 0
Error on sent 1
Directional errors: T1 relaxation makes sent-1 bits fragile.



End-to-end fidelity metrics.